

ICPC Notebook

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template

hash.sh

```
# 使い方: sh hash.sh -> コピペ -> Ctrl + D
# コメント・空白・改行を削除して md5 でハッシュする
g++ -dD -E -P -fpreprocessed - | tr -d '[:space:]' | md5sum |
cut -c-6
```

settings.sh

```
# Clion の設定
Settings → Build → CMake → Reload CMake Project
add_compile_options(-D_GLIBCXX_DEBUG)
# Caps Lock を Ctrl に変更
setxkbmap -option ctrl:nocaps
```

template.hpp

md5: bfa1a2

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;
const ll INF = LLONG_MAX / 4;
template<class T> using V = vector<T>;
#define rep(i, a, b) for(ll i = a; i < (b); i++)
#define each(i, a) for(auto&& i : a)
#define all(a) begin(a), end(a)
#define sz(a) ssize(a)
bool chmin(auto& a, auto b) { return a > b ? a = b, 1 : 0; }
bool chmax(auto& a, auto b) { return a < b ? a = b, 1 : 0; }

int main() {
    cin.tie(0)->sync_with_stdio(0);
    // your code here...
}
```

data-structure

BIT.hpp

md5: ab17a0

```
struct BIT {
    V<ll> a;
    BIT(ll n) : a(n + 1) {}
    void add(ll i, ll x) { // A[i] += x
        i++;
        while(i < sz(a)) {
            a[i] += x;
            i += i & -i;
        }
    }
    ll sum(ll r) {
        ll s = 0;
        while(r) {
            s += a[r];
            r -= r & -r;
        }
        return s;
    }
    ll sum(ll l, ll r) { // sum of A[l, r)
        return sum(r) - sum(l);
    }
};
```

FastSet.hpp

md5: f1f4d2

```
// using u64 = uint64_t;
const u64 B = 64;
struct FastSet {
    u64 n;
    V<V<u64>> a;
    FastSet(u64 n_) : n(n_) {
        do a.emplace_back(n_ = (n_ + B - 1) / B);
        while(n_ > 1);
    }
    // bool operator[](ll i) const { return a[0][i / B] >> (i % B) & 1; }
    void set(ll i) {
        each(v, a) {
            v[i / B] |= 1ULL << (i % B);
            i /= B;
        }
    }
    void reset(ll i) {
```

```
        each(v, a) {
            v[i / B] &= ~(1ULL << (i % B));
            if(v[i / B]) break;
            i /= B;
        }
    }
    ll next(ll i) { // i を超える最小の要素
        rep(h, 0, sz(a)) {
            i++;
            if(i / B >= sz(a[h])) break;
            u64 d = a[h][i / B] >> (i % B);
            if(d) {
                i += countr_zero(d);
                while(h--) i = i * B + countr_zero(a[h][i]);
                return i;
            }
            i /= B;
        }
        return n;
    }
    ll prev(ll i) { // i より小さい最大の要素
        rep(h, 0, sz(a)) {
            i--;
            if(i < 0) break;
            u64 d = a[h][i / B] << (~i % B);
            if(d) {
                i -= countl_zero(d);
                while(h--) i = i * B + __lg(a[h][i]);
                return i;
            }
            i /= B;
        }
        return -1;
    }
};
```

LineContainer.hpp

md5: c4a822

```
struct Line {
    mutable ll a, b, p;
    bool operator<(Line o) const { return a < o.a; }
    bool operator<(ll x) const { return p < x; }
};
// for doubles, use INFINITY and div(a,b) = a/b
struct LineContainer : set<Line, less<>> {
    // floored division (b > 0)
    ll div(ll a, ll b) { return a / b - (a % b < 0); }
    bool check(auto x, auto y) {
        x->p = div(x->b - y->b, y->a - x->a);
        return x->p >= y->p;
    }
    void add(ll a, ll b) { // add line ax + b
        auto [z, f] = emplace(a, b, LLONG_MAX);
        chmax(z->b, b);
        auto y = z++, x = y;
        while(z != end() && check(y, z)) z = erase(z);
        if(x != begin() && check(--x, y)) check(x, y = erase(y));
        while((y = x) != begin() && (--x)->p >= y->p) check(x,
erase(y));
    }
    ll max(ll x) {
        assert(size());
        auto l = *lower_bound(x);
        return l.a * x + l.b;
    }
};
```

math

BinaryGCD.hpp

md5: f3ab31

```
u64 ctz(u64 x) { return countr_zero(x); }
u64 binary_gcd(u64 x, u64 y) {
    if(!x || !y) return x | y;
    u64 n = ctz(x), m = ctz(y);
    x >>= n, y >>= m;
    while(x != y) {
        if(x > y) x = (x - y) >> ctz(x - y);
        else y = (y - x) >> ctz(y - x);
    }
    return x << min(n, m);
}
```

ExtGCD.hpp

md5: c3fa9b

```
// returns gcd(a, b) and assign x, y to integers
// s.t. ax + by = gcd(a, b) and |x| + |y| is minimized
ll extgcd(ll a, ll b, ll& x, ll& y) {
    // assert(a >= 0 && b >= 0);
    if(!b) return x = 1, y = 0, a;
    ll d = extgcd(b, a % b, y, x);
    y -= a / b * x;
    return d;
}
```

modint

BarrettReduction.hpp

md5: 2ca7f3

```
// using u64 = uint64_t;
struct Barrett { // mod < 2^32
    u64 m, im;
    Barrett(u64 mod) : m(mod), im(-1ULL / m + 1) {}
    // input: a * b < 2^64, output: a * b % mod
    u64 mul(u64 a, u64 b) const {
        a *= b;
        u64 x = ((_uint128_t)a * im) >> 64;
        a -= x * m;
        if((ll)a < 0) a += m;
        return a;
    }
};
```

modint.hpp

md5: 81b530

```
const ll mod = 998244353;
struct mm {
    ll x;
    mm(ll x_ = 0) : x(x_ % mod) {
        if(x < 0) x += mod;
    }
    friend mm operator+(mm a, mm b) { return a.x + b.x; }
    friend mm operator-(mm a, mm b) { return a.x - b.x; }
    friend mm operator*(mm a, mm b) { return a.x * b.x; }
    friend mm operator/(mm a, mm b) { return a * b.inv(); }
    // 4 行コピー Alt + Shift + クリックで複数カーソル
    friend mm& operator+=(mm& a, mm b) { return a = a.x + b.x; }
    friend mm& operator-=(mm& a, mm b) { return a = a.x - b.x; }
    friend mm& operator*=(mm& a, mm b) { return a = a.x * b.x; }
    friend mm& operator/=(mm& a, mm b) { return a = a * b.inv(); }
}

mm inv() const { return pow(mod - 2); }
mm pow(ll b) const {
    mm a = *this, c = 1;
    while(b) {
        if(b & 1) c *= a;
        a *= a;
        b >>= 1;
    }
    return c;
}
```

FPS

FFT.hpp

md5: d491f3

```
// {998244353, 3}, {1811939329, 13}, {2013265921, 31}
mm g = 3; // 原始根
void fft(V<mm>& a) {
    ll n = sz(a), lg = __lg(n);
    assert((1 << lg) == n);
    V<mm> b(n);
    rep(l, 1, lg + 1) {
        ll w = n >> l;
        mm s = 1, r = g.pow(mod >> l);
        for(ll u = 0; u < n / 2; u += w) {
            rep(d, 0, w) {
                mm x = a[u << 1 | d], y = a[u << 1 | w | d] * s;
                b[u | d] = x + y;
                b[n >> 1 | u | d] = x - y;
            }
            s *= r;
        }
        swap(a, b);
    }
}
```

```
}
V<mm> conv(V<mm> a, V<mm> b) {
    if(a.empty() || b.empty()) return {};
    size_t s = sz(a) + sz(b) - 1, n = bit_ceil(s);
    // if(min(sz(a), sz(b)) <= 60) 愚直に掛け算
    a.resize(n);
    b.resize(n);
    fft(a);
    fft(b);
    mm inv = mm(n).inv();
    rep(i, 0, n) a[i] *= b[i] * inv;
    reverse(1 + all(a));
    fft(a);
    a.resize(s);
    return a;
}
```

FFT_fast.hpp

md5: 2de3b7

```
// modint を u32 にして加減算を真面目にやると速い
mm g = 3; // 原始根
void fft(V<mm>& a) {
    ll n = sz(a), lg = __lg(n);
    static auto z = [] {
        V<mm> z(30);
        mm s = 1;
        rep(i, 2, 32) {
            z[i - 2] = s * g.pow(mod >> i);
            s *= g.inv().pow(mod >> i);
        }
        return z;
    }();
    rep(l, 0, lg) {
        ll w = 1 << (lg - l - 1);
        mm s = 1;
        rep(k, 0, 1 << l) {
            ll o = k << (lg - l);
            rep(i, o, o + w) {
                mm x = a[i], y = a[i + w] * s;
                a[i] = x + y;
                a[i + w] = x - y;
            }
            s *= z[countr_zero<uint64_t>(~k)];
        }
    }
}
// コピペ
void ifft(V<mm>& a) {
    ll n = sz(a), lg = __lg(n);
    static auto z = [] {
        V<mm> z(30);
        mm s = 1;
        rep(i, 2, 32) { // g を逆数に
            z[i - 2] = s * g.inv().pow(mod >> i);
            s *= g.pow(mod >> i);
        }
        return z;
    }();
    for(ll l = lg; l--;) { // 逆順に
        ll w = 1 << (lg - l - 1);
        mm s = 1;
        rep(k, 0, 1 << l) {
            ll o = k << (lg - l);
            rep(i, o, o + w) {
                mm x = a[i], y = a[i + w]; // *s を下に移動
                a[i] = x + y;
                a[i + w] = (x - y) * s;
            }
            s *= z[countr_zero<uint64_t>(~k)];
        }
    }
}
V<mm> conv(V<mm> a, V<mm> b) {
    if(a.empty() || b.empty()) return {};
    size_t s = sz(a) + sz(b) - 1, n = bit_ceil(s);
    // if(min(sz(a), sz(b)) <= 60) 愚直に掛け算
    a.resize(n);
    b.resize(n);
    fft(a);
    fft(b);
    mm inv = mm(n).inv();
    rep(i, 0, n) a[i] *= b[i] * inv;
    ifft(a);
    a.resize(s);
    return a;
}
```

graph
graph/tree
flow

燃やす埋める.md

変形前の制約	変形後の制約
x が 0 のとき z 失う	(x, T, z)
x が 0 のとき z 得る	無条件で z 得る; (S, x, z)
x が 1 のとき z 失う	(S, x, z)
x が 1 のとき z 得る	無条件で z 得る; (x, T, z)
$x, y, \dots$ がすべて 0 のとき z 得る	無条件で z 得る; $(S, w, z), (w, x, \infty), (w, y, \infty)$
$x, y, \dots$ がすべて 1 のとき z 得る	無条件で z 得る; $(w, T, z), (x, w, \infty), (y, w, \infty)$

string

KMP.hpp

md5: d9caea

```
// kmp[i] := max{ l ≤ i | s[:l] == s[(i+1)-l:i+1] }
// abacaba -> 0010123
auto KMP(string s) {
    V<ll> p(sz(s));
    rep(i, 1, sz(s)) {
        ll g = p[i - 1];
        while(g && s[i] != s[g]) g = p[g - 1];
        p[i] = g + (s[i] == s[g]);
    }
    return p;
}
```

Manacher.hpp

md5: d5b50f

```
// 各位置での回文半径を求める
// aaabaaa -> 1214121
// 偶数長の回文を含めて直径を知るには、N+1 個の $ を挿入して 1 を引く
// $a$a$a$b$a$a$a$ -> 123432181234321
auto manacher(string s) {
    ll n = sz(s), i = 0, j = 0;
    V<ll> r(n);
    while(i < n) {
        while(i >= j && i + j < n && s[i - j] == s[i + j]) j++;
        r[i] = j;
        ll k = 1;
        while(i >= k && i + k < n && k + r[i - k] < j) {
            r[i + k] = r[i - k];
            k++;
        }
        i += k, j -= k;
    }
    return r;
}
```

RollingHash.hpp

md5: f4a7db

```
// using u64 = uint64_t;
const u64 mod = INF;
u64 add(u64 a, u64 b) {
    a += b;
    if(a >= mod) a -= mod;
    return a;
}
u64 mul(u64 a, u64 b) {
    auto c = (__uint128_t)a * b;
    return add(c >> 61, c & mod);
}
const u64 r = random_device{}();
struct RH {
    ll n;
    V<u64> hs, pw;
    RH(string s) : n(sz(s)), hs(n + 1), pw(n + 1, 1) {
        rep(i, 0, n) {
```

```
    }
  }
  u64 get(ll l, ll r) const { return add(hs[r], mod -
mul(hs[l], pw[r - l])); }
};
```

SuffixArray.hpp

md5: ebb633

```
// returns pair{sa, lcp}
// sa 長さ n : s[sa[0]:] < s[sa[1]:] < ... < s[sa[n-1]:]
// lcp 長さ n-1 : lcp[i] = LCP(s[sa[i]:], s[sa[i+1]:])
// O(n log n) time
auto SA(auto s) { // string or vector
  // assert(s.size() >= 1);
  ll n = sz(s);
  V<ll> sa(n), r(n + 1), x(n), y(n + 1), c(n + 1);
  rep(i, 0, n) sa[i] = i;
  ranges::sort(sa, {}, [&](ll i) { return s[i]; });
  r[sa[0]] = 1;
  rep(i, 1, n) r[sa[i]] = r[sa[i - 1]] + (s[sa[i - 1]] !=
s[sa[i]]);
  for(ll k = 1; k < n && r[sa.back()] < n; k *= 2) {
    ll p = 0;
    rep(i, n - k, n) x[p++] = i;
    each(i, sa) if(i >= k) x[p++] = i - k;
    ranges::fill(c, 0);
    each(i, x) c[r[i]]++;
    rep(i, 0, n) c[i + 1] += c[i];
    each(i, x | views::reverse) sa[--c[r[i]]] = i;
    y[sa[0]] = 1;
    rep(i, 1, n) {
      ll a = sa[i - 1], b = sa[i];
      y[b] = y[a] + (r[a] != r[b] || r[min(a + k, n)] !=
r[min(b + k, n)]);
    }
    swap(r, y);
  }
  // if you need lcp array
  x.pop_back();
  ll h = 0;
  rep(i, 0, n) {
    ll p = r[i] - 1;
    if(p == n - 1) {
      h = 0;
      continue;
    }
    ll j = sa[p + 1];
    while(i + h < n && j + h < n && s[i + h] == s[j + h])
h++;
    x[p] = h;
    if(h) h--;
  }
  return pair{sa, x};
}
```

Zalgorithm.hpp

md5: 135912

```
// Z[i] := LCP(s, s[i:])
// abacaba -> 7010301
auto Z(string s) {
  ll n = sz(s), l = -1, r = -1;
  V<ll> z(n, n);
```

```
  rep(i, 1, n) {
    ll& x = z[i] = i < r ? min(r - i, z[i - l]) : 0;
    while(i + x < n && s[i + x] == s[x]) x++;
    if(i + x > r) l = i, r = i + x;
  }
  return z;
}
```

algorithm

GoldenSearch.hpp

md5: 900cf6

```
// return {argmax, max}
// f は [l, r] 上で上に単峰であること
auto golden(double l, double r, auto f) {
  const double p = numbers::phi - 1;
  double r2 = lerp(l, r, p);
  auto fr = f(r2);
  rep(i, 0, 80) { // phi^80 = 5e16
    double l2 = lerp(r, l, p);
    auto fl = f(l2);
    // maximize
    if(fl < fr) l = r, r = l2;
    else r = r2, r2 = l2, fr = fl;
  }
  return pair{r2, fr};
}
```

geometry

memo

Primes.md

素数の個数

n	10^2	10^3	10^4	10^5	10^6	10^7	10^8	10^9	10^{10}
$\pi(n)$	25	168	1229	9592	78498	664579	5.76e+6	5.08e+7	4.55e+8

高度合成数

$\leq n$	10^3	10^4	10^5	10^6	10^7	10^8	10^9
x	840	7560	83160	720720	8648640	73513440	735134400
$d^0(x)$	32	64	128	240	448	768	1344

$\leq n$	10^{10}	10^{11}	10^{12}	10^{13}	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}
$d^0(x)$	2304	4032	6720	10752	17280	26880	41472	64512	103680

素数階乗

n	2	3	5	7	11	13	17	19	23	29
$n\#$	2	6	30	210	2310	30030	510510	9.70e+6	2.23e+8	6.47e+9

階乗

4!	5!	6!	7!	8!	9!	10!	11!	12!	13!
24	120	720	5040	40320	362880	3.63e+6	3.99e+7	4.79e+8	6.23e+9